

Predicting a New Grand Prix

F1 Qualifying Predictor — runbook, Aug 30 2026

Short answer to “is running the prediction from the app enough?”: **no, not for a round the app hasn't seen yet.** Preview and Launch only read already-built practice-session features from `data/processed/features.parquet`. For a brand-new round, two offline scripts must run first — and a third, once qualifying actually happens, so History (and the Prediction page's own scoring) can show the official result.

The 3-step workflow for a new round

Run all three from the repo root, on your machine (not from the app). Replace **13** with the round number you're predicting.

1. Download raw laps (as soon as at least FP1 has run)

```
python scripts/download_2026.py --round 13
```

Pulls that weekend's practice-session laps from FastF1 into `data/raw/2026/`. Safe to re-run — an already-downloaded round is skipped.

2. Build features

```
python scripts/build_features.py --round 13
```

Turns those raw laps into the per-driver feature rows the model actually reads, and rewrites `data/processed/features.parquet`. Run again after FP2/FP3 to refresh with more practice data — a partial weekend (FP1 only) still works, it's just less informed.

3. Predict, in the app

Preview or Launch Prediction now finds this round's features. See “Preview vs. Launch” below for which button to use.

Once qualifying actually happens

A 4th step, run once that round's Q session is over, is what lets History (and a re-Preview of that round on the Prediction page) show the official-result comparison and `Off_Pos` column instead of “Awaiting official result”.

```
python scripts/extract_qualifying_targets.py --round 13
```

Pulls the official Q1/Q2/Q3 times into `data/processed/qualifying_targets.parquet`. Re-run it any time after Q — nothing needs to be re-launched in the app for the comparison to appear; scoring is computed fresh on every page load.

Before Aug 30 2026, all three of these scripts were hardcoded to only ever cover Rounds 1–12 (the training set + the offline test round) — a brand-new round silently had nothing to download or build no matter how many times they ran. The `--round` flag above was added today specifically so a new GP can go through this workflow without a code change each time.

Preview vs. Launch — what's the difference?

Preview (not saved)

Calls the read-only prediction endpoint. Shows the result on screen but writes nothing to disk — it never appears in History, and it never overwrites an existing launch for that round. Safe to click as many times as you want while just looking.

Launch Prediction

Runs the same prediction, then persists it to `data/predictions/launches/` as “the current prediction for this round.” This is what makes a round show up in the History tab, scored automatically once `qualifying_targets.parquet` has its official result. Re-launching an already-launched round overwrites its prior launch (a correction, not a second entry).

Where the app's own “run scripts first” error comes from

Preview/Launch return a 404 ("No features found... Run scripts/download_2026.py + scripts/build_features.py for this round first") whenever features.parquet has no rows for that (year, round). That message is accurate but was previously incomplete on its own for a round beyond 12 — running those two scripts with no arguments wouldn't have covered a new round either, since their default set stopped at Round 12. With the --round flag now in place, the message and the fix finally match.